

Question ID: 2c6f214f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The first term of a sequence is **9**. Each term after the first is **4** times the preceding term. If *w* represents the *n*th term of the sequence, which equation gives *w* in terms of *n*?

Answer

- A. $w = 4(9^n)$
- B. $w = 4(9^{n-1})$
- C. $w = 9(4^n)$
- D. $w = 9(4^{n-1})$

Correct Answer: D

Rationale

Choice D is correct. Since *w* represents the *n*th term of the sequence and **9** is the first term of the sequence, the value of *w* is **9** when the value of *n* is **1**. Since each term after the first is **4** times the preceding term, the value of *w* is **9(4)** when the value of *n* is **2**. Therefore, the value of *w* is **9(4)(4)**, or **9(4)²**, when the value of *n* is **3**. More generally, the value of *w* is **9(4ⁿ⁻¹)** for a given value of *n*. Therefore, the equation $w = 9(4^{n-1})$ gives *w* in terms of *n*.

Choice A is incorrect. This equation describes a sequence for which the first term is **36**, rather than **9**, and each term after the first is **9**, rather than **4**, times the preceding term.

Choice B is incorrect. This equation describes a sequence for which the first term is **4**, rather than **9**, and each term after the first is **9**, rather than **4**, times the preceding term.

Choice C is incorrect. This equation describes a sequence for which the first term is **36**, rather than **9**.